

Total 4 questions in Part A.

Unless given specifically

Take --  $V_{DD} = V_{CC} = 3.3V$ .

For NMOS device  $\mu_n C_{ox} = 140 \mu A/V^2$ ,  $V_T = 0.7V$ ,  $\lambda = 0.1 V^{-1}$ ,  $V_{ov} = 0.2V$

For PMOS device  $\mu_p C_{ox} = 40 \mu A/V^2$ ,  $V_T = -0.7V$ ,  $\lambda = 0.1 V^{-1}$ ,  $V_{ov} = 0.2V$

For NPN/ PNP device  $\beta = 100$ ,  $V_{CE,SAT} = 0.2V$ ,  $V_A = 100V$ ,  $kT/q = 25mV$  (at room temp.),  
 $I_S = 10^{-14} A$ ,  $V_{BE,ON} = 0.5V$ ,  $\alpha \approx 1$

NOTE:

If not specified in question -----

- Ignore  $\gamma$ ,  $\lambda$  in drain current equation.
- Bulk of nmos connected to ground and bulk of pmos connected to  $V_{dd}$ . Specify your assumptions. Justify your answers.
- Unless specified, assume all MOSFETs are biased in the saturation region
- All symbols have the usual meaning

Answer all the sub-parts of a question in sequence and one place only. Clearly show the procedure used to arrive at the answer for full credit. Report the answers with proper units.

Q 1. Consider the amplifier circuit given in Fig. 1. A1 and A2 amplifiers are composed of **only devices**

--Assume all transistors (MOSFET/ BJT) operate in the active region. The current biasing source is a basic current mirror circuit. Assume matching of required components wherever required.

--Assume all N (or P) MOSFETs used in circuits are identical, and all N (or P) BJTs used in circuits (excluding device carrying  $I_{SS}$  current) are identical. Assume the amplifier is operating at a low frequency.

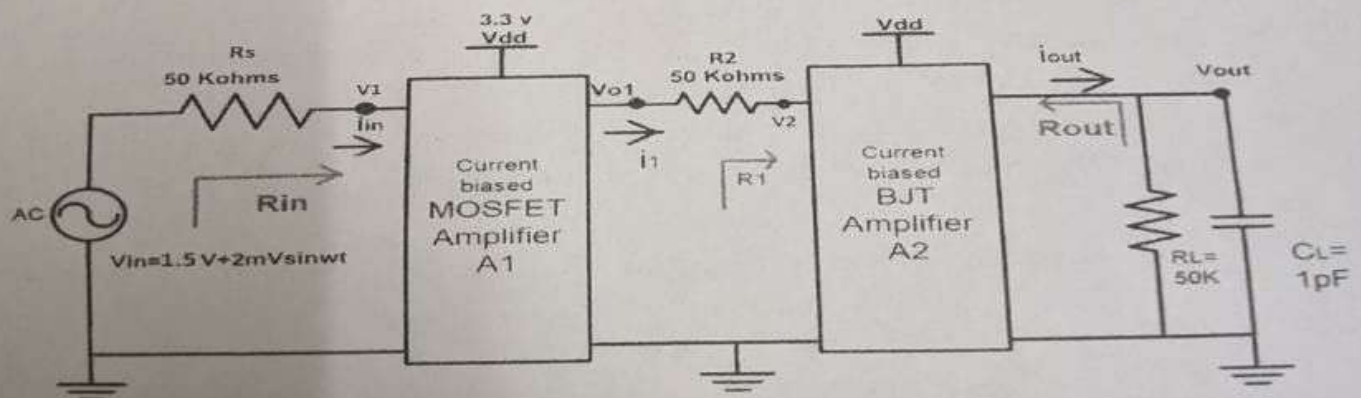


Fig. 1

a) Calculate the values of  $g_m$ ,  $r_o$  of MOSFET device and  $g_m$ ,  $r_o$ ,  $r_\pi$  of BJT device. DC bias current through every MOSFET device  $I_{A1} = 50\mu A$ , and BJT device is  $I_{A2} = 100\mu A$  (excluding  $I_{SS}$  transistor.) Given,  $V_{ov} = 0.2V$ .

b) Sketch and label (neat) total input ( $V_{in}$ ) waveform w.r.t time.

Now, write the answer for parts (c, d, e) if A1 is a common source amplifier and A2 is an emitter follower. Assume the amplifier is designed for maximum output swing.

c) Sketch and label a neat schematic of the entire circuit.

d) Determine the value of  $R_1$ . Compute DC power dissipated in the amplifier.

e) Determine the value of Trans-conductance Gain ( $G_m = i_{out}/V_{in}$ ) of the entire amplifier.

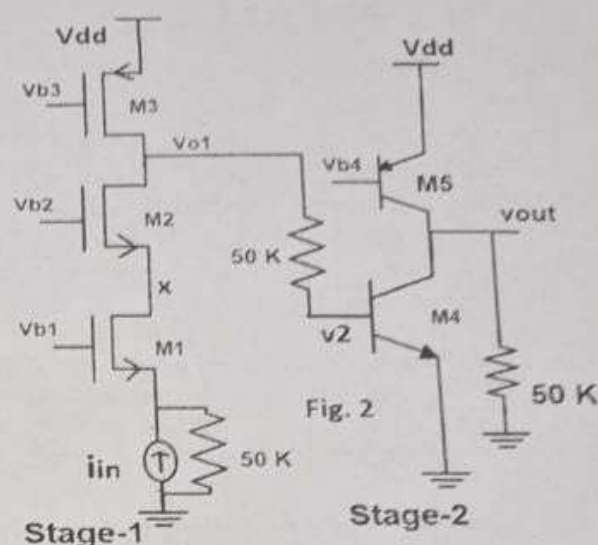
Now, Write the answer for parts (f, g, h) if A1 is a single-ended output differential amplifier with  $v_{id} = v_{in}$ , and A2 is a common base amplifier. Assume the amplifier is designed for maximum output swing.

- f) Sketch and label a neat schematic of the entire circuit
- g) Determine the differential mode gain  $A_{dm} = \frac{v_{o1}}{v_{id}}$ , and common mode gain  $A_{cm} = \frac{v_{o1}}{v_{lcm}}$  for no device mismatch condition.

- h) Determine the magnitude of input offset voltage if a mismatch of components causes a  $V_{o1} = 0.1 \text{ mV}$
- [20 marks]

Q 2. Consider the 2-stage amplifier circuit given in Fig. 2. DC bias current through every MOSFET device is  $50 \mu\text{A}$ , and through BJT device is  $100 \mu\text{A}$ . Assume that given amplifier is designed for maximum output swing

- a) Identify the topology of stage-1 and stage-2 amplifiers in Fig. 2. Also, determine and output voltage signal swing range.
- b) Determine the value of input impedance  $R_{in}$ .
- c) Determine the low-frequency trans-resistance gain  $R_m = \frac{v_{out}}{i_{in}}$ . Hence, sketch and label the trans-resistance amplifier model of the entire amplifier.



[ 10 marks]

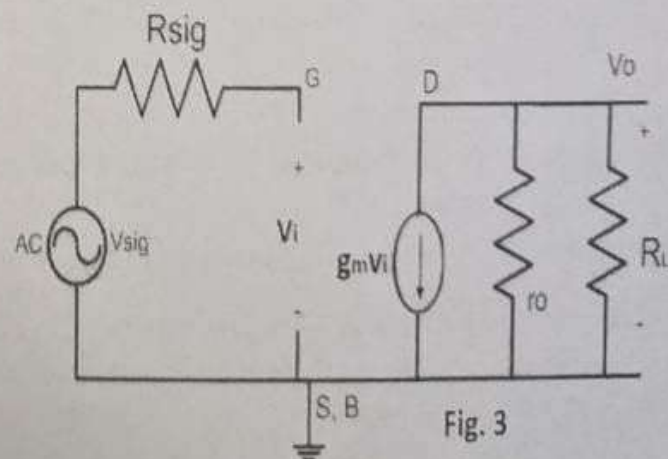
Q 3. Draw a labeled common source amplifier (CSA) circuit with active load. Assume,  $W_{Nmos} = 10 \mu\text{m}$ ,  $W_{Pmos} = 20 \mu\text{m}$ ,  $L_{n,p} = 1 \mu\text{m}$ . Given, overdrive voltage =  $0.2 \text{ V}$ .

- a) Now, use a current mirror to bias (only) active load. Draw the complete circuit (CSA with current mirror active load). Now, calculate the width of transistor/s of the current mirror circuit if the current transfer ratio is 2. Assume  $I_{REF} = 20 \mu\text{A}$ .
- b) Now, modify the circuit of **part (a)** to design a CSA with source degeneration by adding a diode-connected NMOS. Draw and label the complete schematic.
- c) Now for **part (b) circuit**, calculate the low-frequency small-signal voltage gain " $A_v = (v_{out} / v_{in})$ ".
- d) Now for **part (b) circuit**, calculate the voltage gain if the output is tapped from the drain of diode-connected NMOS.
- e) Now, for **part (b)**, investigate the effect of inserting a source degeneration diode-connected NMOS on the current transfer ratio of the current mirror in the circuit. Justify your answer.

[15 marks]

Q 4. A voltage amplifier with AC signal source  $V_{sig}$  and resistance  $R_{sig}$  is shown in Fig. 3. The amplifier has  $V_{dd} = 3.3 \text{ V}$  with a DC current of  $1 \text{ mA}$ .

- a) Find the expression for voltage gain  $(v_o / v_{sig})$ . Now calculate, the value of voltage gain (in dB) if  $r_o = R_L = 1/g_m$ .
- b) Now, sketch and label the voltage transfer characteristics (VTC) of the amplifier if it remains off till  $V_{sig} = 1.2 \text{ V}$  and has a constant magnitude of " $|v_o / v_{sig}|$ " for  $V_{sig} > 1.2 \text{ V}$ .



- c) Calculate power efficiency if  $V_{sig}$  is a sinusoidal voltage of peak  $200 \text{ mV}$  across load  $R_L = 1 \text{ k}\Omega$ .

[15 marks]

-----End of Part-A-----



### Part-B

There are **TWO** questions and each carries **30 Marks**. Answer all the sub-parts of a question in sequence and one place only. Clearly show the procedure used to arrive at the answer for full credit. Report the answers with proper units.

**Question-1:** There are 2 sub-parts in this question and each is independent of the other.

(a) An open-loop amplifier has a gain of **100 dB**, input resistance **100k $\Omega$** ,  **$R_{out} = 0.1k\Omega$** , 3dB bandwidth (single pole) of **1k rad/s**. If it has a closed-loop gain of **40 dB** when shunt-series negative feedback is used. Find the value of feedback factor, loop gain, input and output resistance with feedback, and 3dB bandwidth with feedback. [10]

(b)

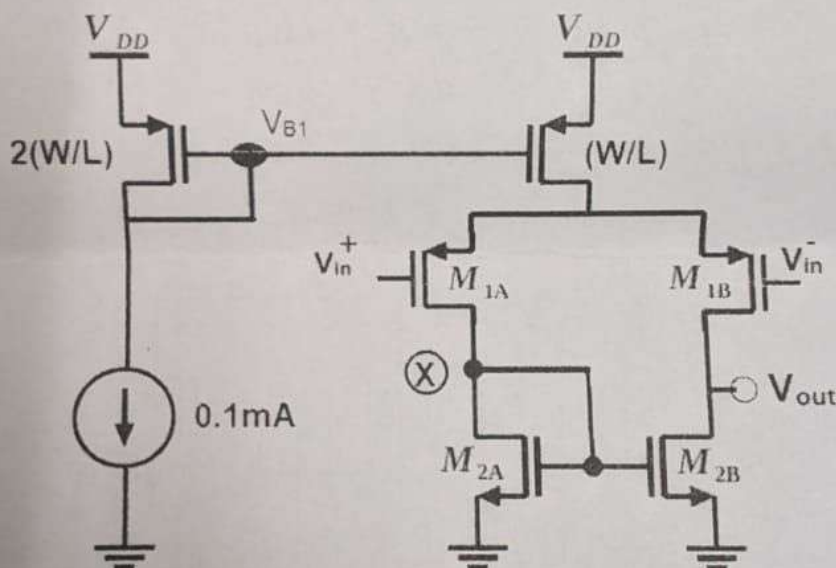


Fig. 1

For the circuit shown in Fig. 1, Assume  $V_{DD} = 2.5\text{ V}$ ,  $V_{ov}$  for each MOSFET is **0.25 V**,  $V_A$  for each MOSFET is **100 V**, and  $V_{th}$  (Threshold voltage) for each MOSFET is **0.65 V**

- Find out the value of differential gain. [2]
- Find the value of  $V_{OUT}$  (voltage Swing). [2]
- Calculate the value of the input common-mode range (ICMR). [3]
- If a differential input of **0.1mV sin $\omega t$**  is applied to the Differential amplifier, write the time domain output voltage equation. Assume, the input is riding on a sufficient DC voltage to make all MOSFETs operate in the active region. [3]
- What is the maximum peak voltage of the input sinusoidal signal that can be faithfully reproduced at the output (without clipping)? [3]
- If the PMOS differential pair transistors have a **5%** mismatch in their  $(W/L)$ , find the input offset voltage of the differential amplifier. [3]
- If  $C_x = 200\text{fF}$  and output  $C_{load} = 2\text{pF}$ , find the location of the dominant pole and mirror pole. [4]

Question-2: There are 3 sub-parts in this question and each is independent of the other.

- (a) The transfer function of an amplifier is given below. Draw the Bode plots for the function. Find the gain margin (in dB) and phase margin (in degrees) and 3-dB bandwidth (in rad/sec). [8]

$$\frac{Y(s)}{X(s)} = \frac{20(s+20)}{(s+2)(s+10)}$$

- (b) Consider the two-stage amplifier shown in Fig. 2. M1, M3 are PMOSFETs, and M2, M4 are NMOSFETs. A voltage source having an internal resistance of  $100k\Omega$  is connected to node P. Use the data given in the Table below wherever necessary for calculations.
- Find the total capacitance at the nodes P and X using Millers' approximation. [6]
  - Using the method of OCTC, find the 3-dB bandwidth of the first stage (STAGE-1). [8]
  - Find the external load capacitance  $C_L$  that needs to be connected at Q so that it contributes a pole at  $500 \text{ krad/sec}$ . [3]

|         | $g_m \text{ (mA/V)}$ | $r_o \text{ (k}\Omega\text{)}$ | $c_{gs} \text{ (fF)}$ | $c_{gd} \text{ (fF)}$ | $c_{sb} \text{ (fF)}$ | $C_{db} \text{ (fF)}$ |
|---------|----------------------|--------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| PMOSFET | 0.2                  | 400                            | 100                   | 50                    | 20                    | 20                    |
| NMOSFET | 0.4                  | 600                            | 120                   | 60                    | 20                    | 20                    |

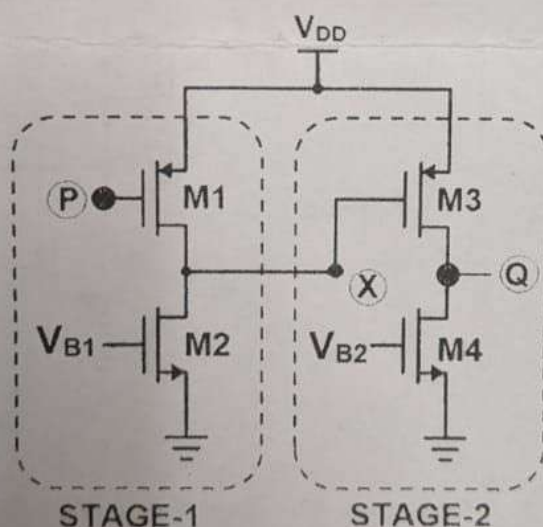


Fig. 2

- (c) A MOSFET differential amplifier is biased with a current source (tail) having an output resistance  $R_{ss} = 100 \text{ k}\Omega$  and an output capacitance  $C_{ss} = 0.1 \text{ pF}$ . If the differential gain is found to have a dominant pole at  $40 \text{ Mrad/sec}$ , what is the 3-dB frequency of the CMRR? [5]

\*\*\* End of Part-B \*\*\*